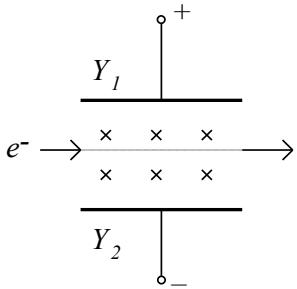
17 An electron is accelerated from rest through a potential difference V (not shown in the diagram). It then passes without any deflection through a region with mutually perpendicular electric and magnetic fields.



The electric field is provided by a pair of deflecting plates Y_1 and Y_2 with potential difference V and separation d. The applied uniform magnetic field is B. The charge to mass ratio of an electron is given by

A $\frac{2B^2d^2}{V}$ **B** $\frac{V}{2B^2d^2}$ **C** $\frac{B^2d^2}{2V}$ **D** $\frac{2V}{B^2d^2}$

18 A, B, C and D are four points on a straight line as shown.